

## Qualification Pack



# Automotive Manufacturing Data Science Specialist

QP Code: ASC/Q6417

Version: 1.0

NSQF Level: 6

Automotive Skills Development Council || 153, GF, Okhla Industrial Area, Phase 3  
New Delhi 110020 || email:ayushi@asdc.org.in

## Qualification Pack

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## Qualification Pack

### ASC/Q6417: Automotive Manufacturing Data Science Specialist

#### Brief Job Description

The individual at this job is responsible for designing and developing the visualization platforms for end-to-end visibility, after-sales analytics solutions and services, and determining repair cost estimates and allocating repair timelines, recording and creating daily maintenance logs and progress reports. He/She leads the development activities and guide the team on the technical front in analytics solutions and reviewing of code and supports a range of analytical, visualization, and predictive modelling projects along with project management.

#### Personal Attributes

The person should have good technical and analytical skills, should have excellent interpersonal skills, communication, and presentation skills, and a good team leader. The person should have project management skills, and also carry out prioritization of work and mentoring the budding engineers

#### Applicable National Occupational Standards (NOS)

##### Compulsory NOS:

1. [ASC/N9810: Manage work and resources \(Manufacturing\)](#)
2. [ASC/N6443: Develop solutions for complex business problems](#)
3. [ASC/N6449: Manage data extraction and collection in automotive manufacturing entities](#)
4. [ASC/N6450: Prepare and analyse data by using analytical tools](#)
5. [ASC/N6451: Analyse assembly line data in automotive manufacturing entity](#)
6. [DGT/VSQ/N0104: Employability Skills \(120 Hours\)](#)

#### Qualification Pack (QP) Parameters

|                   |                        |
|-------------------|------------------------|
| <b>Sector</b>     | Automotive             |
| <b>Sub-Sector</b> | Manufacturing          |
| <b>Occupation</b> | Production Engineering |
| <b>Country</b>    | India                  |
| <b>NSQF Level</b> | 6                      |

## Qualification Pack

|                                                           |                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Credits</b>                                            | 22                                                                                                                                                                                                                                                                                                          |
| <b>Aligned to NCO/ISCO/ISIC Code</b>                      | NCO-2015/2120.0300                                                                                                                                                                                                                                                                                          |
| <b>Minimum Educational Qualification &amp; Experience</b> | B.E./B.Tech with 1 Year of experience in relevant field<br>OR<br>M.E. (Pursuing 2nd year)<br>OR<br>M.Tech (Pursuing 2nd year)<br>OR<br>Certificate-NSQF (Electric Vehicle Product Design Engineer/ Automotive Prototype Manufacturing Lead Technician Level 5) with 3 Years of experience in relevant field |
| <b>Minimum Level of Education for Training in School</b>  |                                                                                                                                                                                                                                                                                                             |
| <b>Pre-Requisite License or Training</b>                  | NA                                                                                                                                                                                                                                                                                                          |
| <b>Minimum Job Entry Age</b>                              | 22 Years                                                                                                                                                                                                                                                                                                    |
| <b>Last Reviewed On</b>                                   | NA                                                                                                                                                                                                                                                                                                          |
| <b>Next Review Date</b>                                   | 23/06/2026                                                                                                                                                                                                                                                                                                  |
| <b>NSQC Approval Date</b>                                 | 23/06/2023                                                                                                                                                                                                                                                                                                  |
| <b>Version</b>                                            | 1.0                                                                                                                                                                                                                                                                                                         |
| <b>Reference code on NQR</b>                              | QG-06-AU-00573-2023-V1-ASDC                                                                                                                                                                                                                                                                                 |
| <b>NQR Version</b>                                        | 1.1                                                                                                                                                                                                                                                                                                         |

## Qualification Pack

### ASC/N9810: Manage work and resources (Manufacturing)

#### Description

This NOS unit is about implementing safety, planning work, adopting sustainable practices for optimising the use of resources.

#### Scope

The scope covers the following :

- Maintain safe and secure working environment
- Maintain Health and Hygiene
- Effective waste management practices
- Material/energy conservation practices

#### Elements and Performance Criteria

##### *Maintain safe and secure working environment*

To be competent, the user/individual on the job must be able to:

- PC1.** identify hazardous activities and the possible causes of risks or accidents in the workplace
- PC2.** implement safe working practices for dealing with hazards to ensure safety of self and others
- PC3.** conduct regular checks of the machines with support of the maintenance team to identify potential hazards
- PC4.** ensure that all the tools/equipment/fasteners/spare parts are arranged as per specifications/utility into proper trays, cabinets, lockers as mentioned in the 5S guidelines/work instructions
- PC5.** organise safety drills or training sessions to create awareness amongst others on the identified risks and safety practices
- PC6.** fill daily check sheet to report improvements done and risks identified
- PC7.** ensure that relevant safety boards/signs are placed on the shop floor for the safety of self and others
- PC8.** report any identified breaches in health, safety and security policies and procedures to the designated person

##### *Maintain Health and Hygiene*

To be competent, the user/individual on the job must be able to:

- PC9.** ensure workplace, equipment, restrooms etc. are sanitized regularly
- PC10.** ensure team is aware about hygiene and sanitation regulations and following them on the shop floor
- PC11.** ensure availability of running water, hand wash and alcohol-based sanitizers at the workplace
- PC12.** report advanced hygiene and sanitation issues to appropriate authority
- PC13.** follow stress and anxiety management techniques and support employees to cope with stress, anxiety etc
- PC14.** wear and dispose PPEs regularly and appropriately

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### *Effective waste management practices*

To be competent, the user/individual on the job must be able to:

**PC15.** ensure recyclable, non-recyclable and hazardous wastes are segregated as per SOP

**PC16.** ensure proper mechanism is followed while collecting and disposing of non-recyclable, recyclable and reusable waste

### *Material/energy conservation practices*

To be competent, the user/individual on the job must be able to:

**PC17.** ensure malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment are resolved effectively

**PC18.** prepare and analyze material and energy audit reports to decipher excessive consumption of material and water

**PC19.** identify possibilities of using renewable energy and environment friendly fuels

**PC20.** identify processes where material and energy/electricity utilization can be optimized

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** organisation procedures for health, safety and security, individual role and responsibilities in this context

**KU2.** the organisation's emergency procedures for different emergency situations and the importance of following the same

**KU3.** evacuation procedures for workers and visitors

**KU4.** how and when to report hazards as well as the limits of responsibility for dealing with hazards

**KU5.** potential hazards, risks and threats based on the nature of work

**KU6.** various types of fire extinguisher

**KU7.** various types of safety signs and their meaning

**KU8.** appropriate first aid treatment relevant to different condition e.g. bleeding, minor burns, eye injuries etc.

**KU9.** relevant standards, procedures and policies related to 5S followed in the company

**KU10.** the various materials used and their storage norms

**KU11.** importance of efficient utilisation of material and water

**KU12.** basics of electricity and prevalent energy efficient devices

**KU13.** common practices of conserving electricity

**KU14.** common sources and ways to minimize pollution

**KU15.** categorisation of waste into dry, wet, recyclable, non-recyclable and items of single-use plastics

**KU16.** waste management techniques

**KU17.** significance of greening

## Generic Skills (GS)

## Qualification Pack

User/individual on the job needs to know how to:

- GS1.** read safety instructions/guidelines
- GS2.** modify work practices to improve them
- GS3.** work with supervisors/team members to carry out work related tasks
- GS4.** complete tasks efficiently and accurately within stipulated time
- GS5.** inform/report to concerned person in case of any problem
- GS6.** make timely decisions for efficient utilization of resources
- GS7.** write reports such as accident report, in at least English/regional language

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                          | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Maintain safe and secure working environment</i>                                                                                                                                                       | <b>20</b>    | <b>13</b>       | -             | <b>8</b>   |
| <b>PC1.</b> identify hazardous activities and the possible causes of risks or accidents in the workplace                                                                                                  | 4            | 2               | -             | 2          |
| <b>PC2.</b> implement safe working practices for dealing with hazards to ensure safety of self and others                                                                                                 | 3            | 1               | -             | 2          |
| <b>PC3.</b> conduct regular checks of the machines with support of the maintenance team to identify potential hazards                                                                                     | 2            | 2               | -             | 1          |
| <b>PC4.</b> ensure that all the tools/equipment/fasteners/spare parts are arranged as per specifications/utility into proper trays, cabinets, lockers as mentioned in the 5S guidelines/work instructions | 3            | 2               | -             | 1          |
| <b>PC5.</b> organise safety drills or training sessions to create awareness amongst others on the identified risks and safety practices                                                                   | 2            | -               | -             | -          |
| <b>PC6.</b> fill daily check sheet to report improvements done and risks identified                                                                                                                       | 2            | 2               | -             | -          |
| <b>PC7.</b> ensure that relevant safety boards/signs are placed on the shop floor for the safety of self and others                                                                                       | 2            | 2               | -             | 1          |
| <b>PC8.</b> report any identified breaches in health, safety and security policies and procedures to the designated person                                                                                | 2            | 2               | -             | 1          |
| <i>Maintain Health and Hygiene</i>                                                                                                                                                                        | <b>13</b>    | <b>7</b>        | -             | <b>5</b>   |
| <b>PC9.</b> ensure workplace, equipment, restrooms etc. are sanitized regularly                                                                                                                           | 3            | 2               | -             | 1          |
| <b>PC10.</b> ensure team is aware about hygiene and sanitation regulations and following them on the shop floor                                                                                           | 2            | 1               | -             | -          |
| <b>PC11.</b> ensure availability of running water, hand wash and alcohol-based sanitizers at the workplace                                                                                                | 2            | 2               | -             | 1          |



### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                          | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC12.</b> report advanced hygiene and sanitation issues to appropriate authority                                                       | 1            | 1               | -             | 1          |
| <b>PC13.</b> follow stress and anxiety management techniques and support employees to cope with stress, anxiety etc                       | 2            | 1               | -             | 1          |
| <b>PC14.</b> wear and dispose PPEs regularly and appropriately                                                                            | 3            | -               | -             | 1          |
| <i>Effective waste management practices</i>                                                                                               | <b>6</b>     | <b>4</b>        | -             | <b>1</b>   |
| <b>PC15.</b> ensure recyclable, non-recyclable and hazardous wastes are segregated as per SOP                                             | 3            | 2               | -             | -          |
| <b>PC16.</b> ensure proper mechanism is followed while collecting and disposing of non-recyclable, recyclable and reusable waste          | 3            | 2               | -             | 1          |
| <i>Material/energy conservation practices</i>                                                                                             | <b>11</b>    | <b>6</b>        | -             | <b>6</b>   |
| <b>PC17.</b> ensure malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment are resolved effectively | 2            | 2               | -             | 1          |
| <b>PC18.</b> prepare and analyze material and energy audit reports to decipher excessive consumption of material and water                | 3            | 2               | -             | 1          |
| <b>PC19.</b> identify possibilities of using renewable energy and environment friendly fuels                                              | 3            | 1               | -             | 2          |
| <b>PC20.</b> identify processes where material and energy/electricity utilization can be optimized                                        | 3            | 1               | -             | 2          |
| <b>NOS Total</b>                                                                                                                          | <b>50</b>    | <b>30</b>       | -             | <b>20</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                           |
|----------------------------|-------------------------------------------|
| <b>NOS Code</b>            | ASC/N9810                                 |
| <b>NOS Name</b>            | Manage work and resources (Manufacturing) |
| <b>Sector</b>              | Automotive                                |
| <b>Sub-Sector</b>          | Generic                                   |
| <b>Occupation</b>          | Generic                                   |
| <b>NSQF Level</b>          | 5                                         |
| <b>Credits</b>             | 2                                         |
| <b>Version</b>             | 2.0                                       |
| <b>Last Reviewed Date</b>  | 08/05/2025                                |
| <b>Next Review Date</b>    | 08/05/2028                                |
| <b>NSQC Clearance Date</b> | 08/05/2025                                |

## Qualification Pack

### ASC/N6443: Develop solutions for complex business problems

#### Description

This NOS unit is about developing machine learning models using the extracted data and making predictive analytics solutions for complex business problems. It also involves project management using project tracking tools and task prioritization for all team members

#### Scope

The scope covers the following :

- Execute training phase in the machine learning project lifecycle
- Execute testing phase in the machine learning project lifecycle
- Deployment of the developed analytics model solution into production
- Deployment of the analytics model solution into production line

#### Elements and Performance Criteria

##### *Execute training phase in the machine learning project lifecycle*

To be competent, the user/individual on the job must be able to:

- PC1.** Select and install relevant libraries and tools for model making
- PC2.** Split and prepare the dataset into training, validation and testing sets
- PC3.** Configure hyperparameters for the selected model, establish the training pipelines and execute the training phase
- PC4.** Store the model and network parameters to be used in the testing phase
- PC5.** Prevent underfitting and overfitting of the model
- PC6.** Solve the imbalanced dataset problem when the samples from minority class are very few
- PC7.** Evaluate the training performance of the machine learning model for training and validation accuracy

##### *Execute testing phase in the machine learning project lifecycle*

To be competent, the user/individual on the job must be able to:

- PC8.** Test the models with testing datasets
- PC9.** Ensure the inference time per sample is as per the business requirement
- PC10.** Evaluate the testing performance of the machine learning model for testing accuracy

##### *Deployment of the developed analytics model solution into production*

To be competent, the user/individual on the job must be able to:

- PC11.** Develop a front-end application to fetch inputs from the user and consume developed model for inference
- PC12.** Verify the production performance of the machine learning model
- PC13.** Analyse performance of the machine learning model and prepare feedback on the wrong predictions
- PC14.** Implement the feedback back to the training phase and retrain the machine learning model

##### *Deployment of the analytics model solution into production line*

## Qualification Pack

To be competent, the user/individual on the job must be able to:

- PC15.** Select relevant libraries and machine learning operations (MLOPS) tools and packages for deploying the analytics model solution into production line
- PC16.** Install the selected libraries and tools for machine learning operations tasks
- PC17.** Setup the continuous training, continuous integration, and continuous delivery pipelines for the developed machine learning models
- PC18.** Monitor the analytics model solution performance in the deployment phase
- PC19.** Carry out commissioning of the end-to-end system

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Organizational policies, procedures, and guidelines that relate to designing and maintaining databases
- KU2.** Organizational policies and procedures for sharing data
- KU3.** Organizational policies and procedures for documenting databases architectures and backup mechanisms
- KU4.** Who to involve while developing the different stages in the machine learning lifecycle
- KU5.** Descriptive and Inferential statistics for creating charts and predictive analytics modelling
- KU6.** Types of data wrangling and data cleaning methods to create visualization
- KU7.** Suitable documentation of the organization for the metadata creation
- KU8.** Updated internal and external network regulations
- KU9.** How to make an API interface for the developed machine learning model in Python using REST API framework
- KU10.** How to diagnose and resolve underfitting, overfitting and imbalanced dataset issues
- KU11.** How to use different machine learning algorithms for specific functions like regression, classification and clustering
- KU12.** How to use python programming constructs for developing machine learning models using open-source libraries like for example, scikit-learn
- KU13.** How to develop necessary front end to consume the developed analytics solution

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** Listen effectively and communicate information accurately
- GS3.** Follow rule-based decision-making processes
- GS4.** Make decisions on suitable courses
- GS5.** Plan and organize the work to achieve targets and meet deadlines
- GS6.** Apply problem-solving approaches to different situations
- GS7.** Analyse the business impact and disseminate relevant information to others

## Qualification Pack

**GS8.** Apply balanced judgments to different situations

**GS9.** Check the work is complete and free from errors

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                              | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Execute training phase in the machine learning project lifecycle</i>                                                       | <b>16</b>    | <b>16</b>       | -             | <b>7</b>   |
| <b>PC1.</b> Select and install relevant libraries and tools for model making                                                  | 2            | 2               | -             | 1          |
| <b>PC2.</b> Split and prepare the dataset into training, validation and testing sets                                          | 3            | 3               | -             | 1          |
| <b>PC3.</b> Configure hyperparameters for the selected model, establish the training pipelines and execute the training phase | 3            | 3               | -             | 1          |
| <b>PC4.</b> Store the model and network parameters to be used in the testing phase                                            | 2            | 2               | -             | 1          |
| <b>PC5.</b> Prevent underfitting and overfitting of the model                                                                 | 2            | 2               | -             | 1          |
| <b>PC6.</b> Solve the imbalanced dataset problem when the samples from minority class are very few                            | 2            | 2               | -             | 1          |
| <b>PC7.</b> Evaluate the training performance of the machine learning model for training and validation accuracy              | 2            | 2               | -             | 1          |
| <i>Execute testing phase in the machine learning project lifecycle</i>                                                        | <b>5</b>     | <b>5</b>        | -             | <b>3</b>   |
| <b>PC8.</b> Test the models with testing datasets                                                                             | 2            | 2               | -             | 1          |
| <b>PC9.</b> Ensure the inference time per sample is as per the business requirement                                           | 1            | 1               | -             | 1          |
| <b>PC10.</b> Evaluate the testing performance of the machine learning model for testing accuracy                              | 2            | 2               | -             | 1          |
| <i>Deployment of the developed analytics model solution into production</i>                                                   | <b>8</b>     | <b>8</b>        | -             | <b>4</b>   |
| <b>PC11.</b> Develop a front-end application to fetch inputs from the user and consume developed model for inference          | 2            | 2               | -             | 1          |
| <b>PC12.</b> Verify the production performance of the machine learning model                                                  | 2            | 2               | -             | 1          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                  | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC13.</b> Analyse performance of the machine learning model and prepare feedback on the wrong predictions                                                      | 2            | 2               | -             | 1          |
| <b>PC14.</b> Implement the feedback back to the training phase and retrain the machine learning model                                                             | 2            | 2               | -             | 1          |
| <i>Deployment of the analytics model solution into production line</i>                                                                                            | <b>11</b>    | <b>11</b>       | -             | <b>6</b>   |
| <b>PC15.</b> Select relevant libraries and machine learning operations (MLOPS) tools and packages for deploying the analytics model solution into production line | 2            | 2               | -             | 1          |
| <b>PC16.</b> Install the selected libraries and tools for machine learning operations tasks                                                                       | 2            | 2               | -             | 1          |
| <b>PC17.</b> Setup the continuous training, continuous integration, and continuous delivery pipelines for the developed machine learning models                   | 3            | 3               | -             | 2          |
| <b>PC18.</b> Monitor the analytics model solution performance in the deployment phase                                                                             | 2            | 2               | -             | 1          |
| <b>PC19.</b> Carry out commissioning of the end-to-end system                                                                                                     | 2            | 2               | -             | 1          |
| <b>NOS Total</b>                                                                                                                                                  | <b>40</b>    | <b>40</b>       | -             | <b>20</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                 |
|----------------------------|-------------------------------------------------|
| <b>NOS Code</b>            | ASC/N6443                                       |
| <b>NOS Name</b>            | Develop solutions for complex business problems |
| <b>Sector</b>              | Automotive                                      |
| <b>Sub-Sector</b>          | Manufacturing                                   |
| <b>Occupation</b>          | Production Engineering                          |
| <b>NSQF Level</b>          | 7                                               |
| <b>Credits</b>             | 6                                               |
| <b>Version</b>             | 1.0                                             |
| <b>Last Reviewed Date</b>  | 23/06/2023                                      |
| <b>Next Review Date</b>    | 23/06/2026                                      |
| <b>NSQC Clearance Date</b> | 23/06/2023                                      |



## Qualification Pack

### ASC/N6449: Manage data extraction and collection in automotive manufacturing entities

#### Description

This NOS unit is about performing tasks related to managing extraction and collection of data from various manufacturing entities

#### Scope

The scope covers the following :

- Monitor assessment of project requirements
- Perform and monitor designing of project outline
- Support in selection of data integration platform to integrate the data from various department

#### Elements and Performance Criteria

##### *Monitor assessment of project requirements*

To be competent, the user/individual on the job must be able to:

- PC1.** Support during evaluation of the project requirements to be catered with either visualization platforms or analytics and predictive modelling solutions
- PC2.** Monitor and guide team during designing of data architecture for collection and extraction of data from various departments using connectors and platforms
- PC3.** Identify the people required to execute the business analytics project requirements
- PC4.** Assess all organizational processes related to the use of data and analytics
- PC5.** Check that the existing setup is capable or not for data collection and analysis

##### *Perform and monitor designing of project outline*

To be competent, the user/individual on the job must be able to:

- PC6.** Plan and prepare project layout where it should defend the choice of technology and its cost
- PC7.** Support and prepare the outline of the development process and its requirements for both material and resources
- PC8.** Define various data attributes and what level of analytics is required to analyse data and deliver value
- PC9.** Prepare the timeline and resource requirements
- PC10.** Manage project by using appropriate project tracking tools and task prioritization for all team members
- PC11.** Obtain the necessary approvals within the organization for data collection and extraction from various departments

##### *Support in selection of data integration platform to integrate the data from various department*

To be competent, the user/individual on the job must be able to:

- PC12.** Select the data integration platform with the capabilities like- data transformation, application connectors, file processing, routing, orchestration, event handling, stream processing, API management, no-vendor lock-in.

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- PC13.** Create and monitor an end-to-end data flow using ETL (Extract-Transform-Load) tool using different connectors for different types of data sources
- PC14.** Design and create a data warehouse for data acquisition
- PC15.** Develop data pipelines using connectors to populate the data in the data warehouse

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Organizational policies, procedures, and guidelines that relate to designing and maintaining databases
- KU2.** Organizational policies and procedures for sharing data
- KU3.** Organizational policies and procedures for documenting databases architectures and backup mechanisms
- KU4.** Who to involve while designing and developing the database architecture and pipelines for the solution
- KU5.** Range of standard platforms and tools available and how to use them
- KU6.** Database connectors and application connectors for application-cloud communications
- KU7.** Updated internal and external cybersecurity regulations
- KU8.** Impacts of network on the environment and human health
- KU9.** ETL tools like Talend, SQL Server Integration Services (SSIS), etc
- KU10.** Basics of SQL

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** Listen effectively and communicate information accurately
- GS3.** Follow rule-based decision-making processes
- GS4.** Make decisions on suitable courses
- GS5.** Plan and organize the work to achieve targets and meet deadlines
- GS6.** Apply problem-solving approaches to different situations
- GS7.** Analyse the business impact and disseminate relevant information to others
- GS8.** Apply balanced judgments to different situations
- GS9.** Check the work is complete and free from errors

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                       | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Monitor assessment of project requirements</i>                                                                                                                      | <b>11</b>    | <b>11</b>       | -             | <b>5</b>   |
| <b>PC1.</b> Support during evaluation of the project requirements to be catered with either visualization platforms or analytics and predictive modelling solutions    | 3            | 3               | -             | 1          |
| <b>PC2.</b> Monitor and guide team during designing of data architecture for collection and extraction of data from various departments using connectors and platforms | 2            | 2               | -             | 1          |
| <b>PC3.</b> Identify the people required to execute the business analytics project requirements                                                                        | 2            | 2               | -             | 1          |
| <b>PC4.</b> Assess all organizational processes related to the use of data and analytics                                                                               | 2            | 2               | -             | 1          |
| <b>PC5.</b> Check that the existing setup is capable or not for data collection and analysis                                                                           | 2            | 2               | -             | 1          |
| <i>Perform and monitor designing of project outline</i>                                                                                                                | <b>20</b>    | <b>20</b>       | -             | <b>11</b>  |
| <b>PC6.</b> Plan and prepare project layout where it should defend the choice of technology and its cost                                                               | 4            | 4               | -             | 2          |
| <b>PC7.</b> Support and prepare the outline of the development process and its requirements for both material and resources                                            | 3            | 3               | -             | 2          |
| <b>PC8.</b> Define various data attributes and what level of analytics is required to analyse data and deliver value                                                   | 3            | 3               | -             | 2          |
| <b>PC9.</b> Prepare the timeline and resource requirements                                                                                                             | 3            | 3               | -             | 2          |
| <b>PC10.</b> Manage project by using appropriate project tracking tools and task prioritization for all team members                                                   | 4            | 4               | -             | 2          |
| <b>PC11.</b> Obtain the necessary approvals within the organization for data collection and extraction from various departments                                        | 3            | 3               | -             | 1          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                                                                          | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Support in selection of data integration platform to integrate the data from various department</i>                                                                                                                                    | 9            | 9               | -             | 4          |
| <b>PC12.</b> Select the data integration platform with the capabilities like- data transformation, application connectors, file processing, routing, orchestration, event handling, stream processing, API management, no-vendor lock-in. | 2            | 2               | -             | 1          |
| <b>PC13.</b> Create and monitor an end-to-end data flow using ETL (Extract-Transform-Load) tool using different connectors for different types of data sources                                                                            | 3            | 3               | -             | 1          |
| <b>PC14.</b> Design and create a data warehouse for data acquisition                                                                                                                                                                      | 2            | 2               | -             | 1          |
| <b>PC15.</b> Develop data pipelines using connectors to populate the data in the data warehouse                                                                                                                                           | 2            | 2               | -             | 1          |
| <b>NOS Total</b>                                                                                                                                                                                                                          | <b>40</b>    | <b>40</b>       | <b>-</b>      | <b>20</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                            |
|----------------------------|----------------------------------------------------------------------------|
| <b>NOS Code</b>            | ASC/N6449                                                                  |
| <b>NOS Name</b>            | Manage data extraction and collection in automotive manufacturing entities |
| <b>Sector</b>              | Automotive                                                                 |
| <b>Sub-Sector</b>          | Manufacturing                                                              |
| <b>Occupation</b>          | Production Engineering                                                     |
| <b>NSQF Level</b>          | 6                                                                          |
| <b>Credits</b>             | 2                                                                          |
| <b>Version</b>             | 1.0                                                                        |
| <b>Last Reviewed Date</b>  | 23/06/2023                                                                 |
| <b>Next Review Date</b>    | 23/06/2026                                                                 |
| <b>NSQC Clearance Date</b> | 23/06/2023                                                                 |

## Qualification Pack

### ASC/N6450: Prepare and analyse data by using analytical tools

#### Description

This NOS unit is about performing exploratory data analysis on the data extracted, deciding which data attributes are required for analytics and analysing the extracted attributes using excel/ open-source python libraires

#### Scope

The scope covers the following :

- Identify business goals for which data need to be analysed,
- Support in preparation of data
- Perform statistical analysis of data

#### Elements and Performance Criteria

##### *Identify business goal for which data need to be analysed*

To be competent, the user/individual on the job must be able to:

- PC1.** Define business problems and business goals which can be achieved using available datasets from the manager
- PC2.** Select the relevant source of data to define business goal
- PC3.** Validate the criterion in the business problem with domain person

##### *Support in preparation of data*

To be competent, the user/individual on the job must be able to:

- PC4.** Support and create a set of metadata for the selected dataset
- PC5.** Identify the attributes or columns in the datasets which are most significant from analysis perspective
- PC6.** Perform exploratory data analysis to check for missing or duplicate data

##### *Perform statistical analysis of data*

To be competent, the user/individual on the job must be able to:

- PC7.** Perform descriptive statistical analysis on the data by following SOP
- PC8.** Perform inferential statistics analysis on the data by following SOP
- PC9.** Prepare list of highly correlated attributes
- PC10.** Find correlation amongst the selected attributes of the data and plot their heatmap

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Organizational policies, procedures, and guidelines that relate to designing and maintaining databases
- KU2.** Organizational policies and procedures for sharing data

## Qualification Pack

- KU3.** Organizational policies and procedures for documenting databases architectures and backup mechanisms
- KU4.** Descriptive and Inferential statistics for creating charts and predictive analytics modelling
- KU5.** Types of data wrangling and data cleaning methods to create visualization
- KU6.** Suitable documentation of the organization for the metadata creation
- KU7.** Aggregate the charts to create a dashboard to address the business problem
- KU8.** Addition of filters and chart tips to make the dashboard interactive

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** Listen effectively and communicate information accurately
- GS3.** Follow rule-based decision-making processes
- GS4.** Make decisions on suitable courses
- GS5.** Plan and organize the work to achieve targets and meet deadlines
- GS6.** Apply problem-solving approaches to different situations
- GS7.** Analyse the business impact and disseminate relevant information to others
- GS8.** Apply balanced judgments to different situations
- GS9.** Check the work is complete and free from errors

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                        | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Identify business goal for which data need to be analysed</i>                                                        | <b>11</b>    | <b>11</b>       | -             | <b>6</b>   |
| <b>PC1.</b> Define business problems and business goals which can be achieved using available datasets from the manager | 3            | 3               | -             | 2          |
| <b>PC2.</b> Select the relevant source of data to define business goal                                                  | 4            | 4               | -             | 2          |
| <b>PC3.</b> Validate the criterion in the business problem with domain person                                           | 4            | 4               | -             | 2          |
| <i>Support in preparation of data</i>                                                                                   | <b>16</b>    | <b>16</b>       | -             | <b>7</b>   |
| <b>PC4.</b> Support and create a set of metadata for the selected dataset                                               | 6            | 6               | -             | 3          |
| <b>PC5.</b> Identify the attributes or columns in the datasets which are most significant from analysis perspective     | 5            | 5               | -             | 2          |
| <b>PC6.</b> Perform exploratory data analysis to check for missing or duplicate data                                    | 5            | 5               | -             | 2          |
| <i>Perform statistical analysis of data</i>                                                                             | <b>13</b>    | <b>13</b>       | -             | <b>7</b>   |
| <b>PC7.</b> Perform descriptive statistical analysis on the data by following SOP                                       | 3            | 3               | -             | 2          |
| <b>PC8.</b> Perform inferential statistics analysis on the data by following SOP                                        | 3            | 3               | -             | 2          |
| <b>PC9.</b> Prepare list of highly correlated attributes                                                                | 3            | 3               | -             | 1          |
| <b>PC10.</b> Find correlation amongst the selected attributes of the data and plot their heatmap                        | 4            | 4               | -             | 2          |
| <b>NOS Total</b>                                                                                                        | <b>40</b>    | <b>40</b>       | -             | <b>20</b>  |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                    |
|----------------------------|----------------------------------------------------|
| <b>NOS Code</b>            | ASC/N6450                                          |
| <b>NOS Name</b>            | Prepare and analyse data by using analytical tools |
| <b>Sector</b>              | Automotive                                         |
| <b>Sub-Sector</b>          | Manufacturing                                      |
| <b>Occupation</b>          | Production Engineering                             |
| <b>NSQF Level</b>          | 6                                                  |
| <b>Credits</b>             | 2                                                  |
| <b>Version</b>             | 1.0                                                |
| <b>Last Reviewed Date</b>  | 23/06/2023                                         |
| <b>Next Review Date</b>    | 23/06/2026                                         |
| <b>NSQC Clearance Date</b> | 23/06/2023                                         |

## Qualification Pack

### ASC/N6451: Analyse assembly line data in automotive manufacturing entity

#### Description

This NOS unit is about performing analysis projects related to production process in assembly line using analytics and Business Intelligence tools.

#### Scope

The scope covers the following :

- Optimization of production processes to improve output and efficiency
- Identification of data needs expert analysis
- Deployment of the projects in local server or on the cloud

#### Elements and Performance Criteria

##### *Optimization of production processes to improve output and efficiency*

To be competent, the user/individual on the job must be able to:

- PC1.** Create demand forecast of the project to ensure the right spare parts mix in assembly line
- PC2.** Analyse safety and quality data to reduce risk in assembly line
- PC3.** Conduct predictive health maintenance of assembly line machines

##### *Identification of data needs expert analysis*

To be competent, the user/individual on the job must be able to:

- PC4.** Support in identification and verification of sensor level data sources in assembly line
- PC5.** Validate the data which needs expert analysis

##### *Deployment of the projects in local server or on the cloud*

To be competent, the user/individual on the job must be able to:

- PC6.** Support in deployment of the project on the local server or cloud
- PC7.** Monitor and verify the compatibility of dashboard on different devices
- PC8.** Monitor the alert system in real time dashboard as per requirement

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Organizational policies, procedures, and guidelines that relate to designing and maintaining networks
- KU2.** Organizational policies and procedures for sharing data
- KU3.** Organizational policies and procedures for documenting network designs and fall-back mechanisms
- KU4.** Different types of visualizations charts Bar Graph, Line Graph, Stacked Bar Graph, Pie Chart, Scatter Plot Chart, etc.

## Qualification Pack

- KU5.** Different types and categories of data variables qualitative, quantitative, nominal, ordinal, discrete, continuous, etc
- KU6.** Different types of visualizations tools like Microsoft Power BI Desktop, Tableau Public
- KU7.** Local machine server architecture
- KU8.** Python based on tools like Anaconda, Jupyter, VS Code, etc.

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** Listen effectively and communicate information accurately
- GS3.** Follow rule-based decision-making processes
- GS4.** Make decisions on suitable courses
- GS5.** Plan and organize the work to achieve targets and meet deadlines
- GS6.** Refer anomalies to the supervisor
- GS7.** Ask for clarification and advice from appropriate people
- GS8.** Analyse the business impact and disseminate relevant information to others
- GS9.** Apply balanced judgments to different situations
- GS10.** Check the work is complete and free from errors
- GS11.** Work independently and collaboratively

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                       | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Optimization of production processes to improve output and efficiency</i>                           | <b>16</b>    | <b>16</b>       | -             | <b>8</b>   |
| <b>PC1.</b> Create demand forecast of the project to ensure the right spare parts mix in assembly line | 5            | 5               | -             | 3          |
| <b>PC2.</b> Analyse safety and quality data to reduce risk in assembly line                            | 5            | 5               | -             | 2          |
| <b>PC3.</b> Conduct predictive health maintenance of assembly line machines                            | 6            | 6               | -             | 3          |
| <i>Identification of data needs expert analysis</i>                                                    | <b>11</b>    | <b>11</b>       | -             | <b>5</b>   |
| <b>PC4.</b> Support in identification and verification of sensor level data sources in assembly line   | 6            | 6               | -             | 3          |
| <b>PC5.</b> Validate the data which needs expert analysis                                              | 5            | 5               | -             | 2          |
| <i>Deployment of the projects in local server or on the cloud</i>                                      | <b>13</b>    | <b>13</b>       | -             | <b>7</b>   |
| <b>PC6.</b> Support in deployment of the project on the local server or cloud                          | 4            | 4               | -             | 2          |
| <b>PC7.</b> Monitor and verify the compatibility of dashboard on different devices                     | 5            | 5               | -             | 3          |
| <b>PC8.</b> Monitor the alert system in real time dashboard as per requirement                         | 4            | 4               | -             | 2          |
| <b>NOS Total</b>                                                                                       | <b>40</b>    | <b>40</b>       | -             | <b>20</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                               |
|----------------------------|---------------------------------------------------------------|
| <b>NOS Code</b>            | ASC/N6451                                                     |
| <b>NOS Name</b>            | Analyse assembly line data in automotive manufacturing entity |
| <b>Sector</b>              | Automotive                                                    |
| <b>Sub-Sector</b>          | Manufacturing                                                 |
| <b>Occupation</b>          | Production Engineering                                        |
| <b>NSQF Level</b>          | 6                                                             |
| <b>Credits</b>             | 2                                                             |
| <b>Version</b>             | 1.0                                                           |
| <b>Last Reviewed Date</b>  | 23/06/2023                                                    |
| <b>Next Review Date</b>    | 23/06/2026                                                    |
| <b>NSQC Clearance Date</b> | 23/06/2023                                                    |

## Qualification Pack

### DGT/VSQ/N0104: Employability Skills (120 Hours)

#### Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

#### Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

#### Elements and Performance Criteria

##### *Introduction to Employability Skills*

To be competent, the user/individual on the job must be able to:

- PC1.** understand the significance of employability skills in meeting the current job market requirement and future of work
- PC2.** identify and explore learning and employability relevant portals
- PC3.** research about the different industries, job market trends, latest skills required and the available opportunities

##### *Constitutional values - Citizenship*

To be competent, the user/individual on the job must be able to:

- PC4.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. for personal growth and the nation's progress
- PC5.** follow personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC6.** follow and promote environmentally sustainable practices

##### *Becoming a Professional in the 21st Century*

To be competent, the user/individual on the job must be able to:

- PC7.** recognize the significance of 21st Century Skills for employment

## Qualification Pack

- PC8.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life
- PC9.** adopt a continuous learning mindset for personal and professional development

### *Basic English Skills*

To be competent, the user/individual on the job must be able to:

- PC10.** use English as a medium of formal and informal communication while dealing with topics of everyday conversation in different contexts
- PC11.** speak over the phone in English, in an audible manner, using appropriate greetings, opening, and closing statements both on personal and work front
- PC12.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC13.** write short messages, notes, letters, e-mails etc., using accurate English

### *Career Development & Goal Setting*

To be competent, the user/individual on the job must be able to:

- PC14.** identify career goals based on the skills, interests, knowledge, and personal attributes
- PC15.** prepare a career development plan with short- and long-term goals

### *Communication Skills*

To be competent, the user/individual on the job must be able to:

- PC16.** follow verbal and non-verbal communication etiquette while communicating in professional and public settings
- PC17.** use active listening techniques for effective communication
- PC18.** communicate in writing using appropriate style and format based on formal or informal requirements
- PC19.** work collaboratively with others in a team

### *Diversity & Inclusion*

To be competent, the user/individual on the job must be able to:

- PC20.** • ensure personal behaviour, conduct, and use appropriate communication by taking gender into consideration
- PC21.** empathize with a PwD and aid a PwD, if asked
- PC22.** escalate any issues related to sexual harassment at the workplace in accordance with the POSH Act

### *Financial and Legal Literacy*

To be competent, the user/individual on the job must be able to:

- PC23.** identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.
- PC24.** carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook
- PC25.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC26.** identify relevant rights and laws and use legal aids to fight against legal exploitation

### *Essential Digital Skills*

## Qualification Pack

To be competent, the user/individual on the job must be able to:

- PC27.** operate digital devices and use their features and applications securely and safely
- PC28.** carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.
- PC29.** display responsible online behaviour while using various social media platforms
- PC30.** create a personal email account, send and process received messages as per requirement
- PC31.** carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications
- PC32.** utilize virtual collaboration tools to work effectively

### Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC33.** identify different types of Entrepreneurship and Enterprises
- PC34.** use research and networking skills to identify and assess opportunities for potential business
- PC35.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC36.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

### Customer Service

To be competent, the user/individual on the job must be able to:

- PC37.** identify different types of customers
- PC38.** identify and respond to customer requests and needs in a professional manner
- PC39.** use appropriate tools to collect customer feedback
- PC40.** follow appropriate hygiene and grooming standards

### Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

- PC41.** create a professional Curriculum vitae (Résumé)
- PC42.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively
- PC43.** apply to identified job openings using offline /online methods as per requirement
- PC44.** answer questions politely, with clarity and confidence, during recruitment and selection
- PC45.** identify apprenticeship opportunities and register for it as per guidelines and requirements

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** need for employability skills
- KU2.** different learning and employability related portals
- KU3.** various constitutional and personal values
- KU4.** different environmentally sustainable practices and their importance
- KU5.** Twenty first (21st) century skills and their importance



## Qualification Pack

- KU6.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up
- KU7.** importance of career development and setting long- and short-term goals
- KU8.** Do's and don'ts of effective communication
- KU9.** POSH Act
- KU10.** inclusivity and its importance
- KU11.** different types of disabilities and appropriate verbal and non-verbal communication and behaviour towards PwD
- KU12.** different types of financial institutes, products, and services
- KU13.** components of salary and how to compute income and expenditure
- KU14.** importance of maintaining safety and security in offline and online financial transactions
- KU15.** different legal rights and laws
- KU16.** different types of digital devices and the procedure to operate them safely and securely
- KU17.** how to create and operate an e- mail account
- KU18.** use applications such as word processors, spreadsheets etc.
- KU19.** different types of Enterprises and ways to identify business opportunities
- KU20.** types and needs of customers
- KU21.** how to apply for a job and prepare for an interview
- KU22.** apprenticeship scheme and the process of registering on apprenticeship portal

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and write different types of documents/instructions/correspondence in English and other languages
- GS2.** communicate effectively using appropriate language in formal and informal settings
- GS3.** behave politely and appropriately with all to maintain effective work relationship
- GS4.** how to work in a virtual mode, using various technological platforms
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Introduction to Employability Skills</i>                                                                                                                                                                                                                                                                      | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC1.</b> understand the significance of employability skills in meeting the current job market requirement and future of work                                                                                                                                                                                 | -            | -               | -             | -          |
| <b>PC2.</b> identify and explore learning and employability relevant portals                                                                                                                                                                                                                                     | -            | -               | -             | -          |
| <b>PC3.</b> research about the different industries, job market trends, latest skills required and the available opportunities                                                                                                                                                                                   | -            | -               | -             | -          |
| <i>Constitutional values – Citizenship</i>                                                                                                                                                                                                                                                                       | <b>2</b>     | <b>1</b>        | -             | -          |
| <b>PC4.</b> recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. for personal growth and the nation's progress                                                                                                               | -            | -               | -             | -          |
| <b>PC5.</b> follow personal values and ethics such as honesty, integrity, caring and respecting others, etc.                                                                                                                                                                                                     | -            | -               | -             | -          |
| <b>PC6.</b> follow and promote environmentally sustainable practices                                                                                                                                                                                                                                             | -            | -               | -             | -          |
| <i>Becoming a Professional in the 21st Century</i>                                                                                                                                                                                                                                                               | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC7.</b> recognize the significance of 21st Century Skills for employment                                                                                                                                                                                                                                     | -            | -               | -             | -          |
| <b>PC8.</b> practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life | -            | -               | -             | -          |
| <b>PC9.</b> adopt a continuous learning mindset for personal and professional development                                                                                                                                                                                                                        | -            | -               | -             | -          |
| <i>Basic English Skills</i>                                                                                                                                                                                                                                                                                      | <b>2</b>     | <b>3</b>        | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                         | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> use English as a medium of formal and informal communication while dealing with topics of everyday conversation in different contexts                       | -            | -               | -             | -          |
| <b>PC11.</b> speak over the phone in English, in an audible manner, using appropriate greetings, opening, and closing statements both on personal and work front         | -            | -               | -             | -          |
| <b>PC12.</b> read and understand routine information, notes, instructions, mails, letters etc. written in English                                                        | -            | -               | -             | -          |
| <b>PC13.</b> write short messages, notes, letters, e-mails etc., using accurate English                                                                                  | -            | -               | -             | -          |
| <i>Career Development &amp; Goal Setting</i>                                                                                                                             | <b>1</b>     | <b>2</b>        | -             | -          |
| <b>PC14.</b> identify career goals based on the skills, interests, knowledge, and personal attributes                                                                    | -            | -               | -             | -          |
| <b>PC15.</b> prepare a career development plan with short- and long-term goals                                                                                           | -            | -               | -             | -          |
| <i>Communication Skills</i>                                                                                                                                              | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC16.</b> follow verbal and non-verbal communication etiquette while communicating in professional and public settings                                                | -            | -               | -             | -          |
| <b>PC17.</b> use active listening techniques for effective communication                                                                                                 | -            | -               | -             | -          |
| <b>PC18.</b> communicate in writing using appropriate style and format based on formal or informal requirements                                                          | -            | -               | -             | -          |
| <b>PC19.</b> work collaboratively with others in a team                                                                                                                  | -            | -               | -             | -          |
| <i>Diversity &amp; Inclusion</i>                                                                                                                                         | <b>1</b>     | <b>2</b>        | -             | -          |
| <b>PC20.</b> <ul style="list-style-type: none"> <li>ensure personal behaviour, conduct, and use appropriate communication by taking gender into consideration</li> </ul> | -            | -               | -             | -          |
| <b>PC21.</b> empathize with a PwD and aid a PwD, if asked                                                                                                                | -            | -               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                     | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC22.</b> escalate any issues related to sexual harassment at the workplace in accordance with the POSH Act                                                                       | -            | -               | -             | -          |
| <i>Financial and Legal Literacy</i>                                                                                                                                                  | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC23.</b> identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.               | -            | -               | -             | -          |
| <b>PC24.</b> carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook                                   | -            | -               | -             | -          |
| <b>PC25.</b> identify common components of salary and compute income, expenses, taxes, investments etc                                                                               | -            | -               | -             | -          |
| <b>PC26.</b> identify relevant rights and laws and use legal aids to fight against legal exploitation                                                                                | -            | -               | -             | -          |
| <i>Essential Digital Skills</i>                                                                                                                                                      | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC27.</b> operate digital devices and use their features and applications securely and safely                                                                                     | -            | -               | -             | -          |
| <b>PC28.</b> carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc. | -            | -               | -             | -          |
| <b>PC29.</b> display responsible online behaviour while using various social media platforms                                                                                         | -            | -               | -             | -          |
| <b>PC30.</b> create a personal email account, send and process received messages as per requirement                                                                                  | -            | -               | -             | -          |
| <b>PC31.</b> carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications                                                   | -            | -               | -             | -          |
| <b>PC32.</b> utilize virtual collaboration tools to work effectively                                                                                                                 | -            | -               | -             | -          |
| <i>Entrepreneurship</i>                                                                                                                                                              | <b>2</b>     | <b>3</b>        | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC33.</b> identify different types of Entrepreneurship and Enterprises                                                                                                        | -            | -               | -             | -          |
| <b>PC34.</b> use research and networking skills to identify and assess opportunities for potential business                                                                      | -            | -               | -             | -          |
| <b>PC35.</b> develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion                                                      | -            | -               | -             | -          |
| <b>PC36.</b> identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity                                           | -            | -               | -             | -          |
| <i>Customer Service</i>                                                                                                                                                          | <b>1</b>     | <b>2</b>        | -             | -          |
| <b>PC37.</b> identify different types of customers                                                                                                                               | -            | -               | -             | -          |
| <b>PC38.</b> identify and respond to customer requests and needs in a professional manner                                                                                        | -            | -               | -             | -          |
| <b>PC39.</b> use appropriate tools to collect customer feedback                                                                                                                  | -            | -               | -             | -          |
| <b>PC40.</b> follow appropriate hygiene and grooming standards                                                                                                                   | -            | -               | -             | -          |
| <i>Getting ready for apprenticeship &amp; Jobs</i>                                                                                                                               | <b>2</b>     | <b>4</b>        | -             | -          |
| <b>PC41.</b> create a professional Curriculum vitae (Résumé)                                                                                                                     | -            | -               | -             | -          |
| <b>PC42.</b> search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively | -            | -               | -             | -          |
| <b>PC43.</b> apply to identified job openings using offline /online methods as per requirement                                                                                   | -            | -               | -             | -          |
| <b>PC44.</b> answer questions politely, with clarity and confidence, during recruitment and selection                                                                            | -            | -               | -             | -          |
| <b>PC45.</b> identify apprenticeship opportunities and register for it as per guidelines and requirements                                                                        | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                 | <b>20</b>    | <b>30</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                  |
|----------------------------|----------------------------------|
| <b>NOS Code</b>            | DGT/VSQ/N0104                    |
| <b>NOS Name</b>            | Employability Skills (120 Hours) |
| <b>Sector</b>              | Cross Sectoral                   |
| <b>Sub-Sector</b>          | Professional Skills              |
| <b>Occupation</b>          | Employability                    |
| <b>NSQF Level</b>          | 6                                |
| <b>Credits</b>             | 4                                |
| <b>Version</b>             | 1.0                              |
| <b>Last Reviewed Date</b>  | 07/10/2025                       |
| <b>Next Review Date</b>    | 07/10/2028                       |
| <b>NSQC Clearance Date</b> | 07/10/2025                       |

## Assessment Guidelines and Assessment Weightage

### Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training centre (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training centre based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.
6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

### Minimum Aggregate Passing % at QP Level : 70

## Qualification Pack

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

### Assessment Weightage

Compulsory NOS

| National Occupational Standards                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage  |
|--------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|------------|
| ASC/N9810.Manage work and resources (Manufacturing)                                  | 50           | 30              | -             | 20         | 100         | 15         |
| ASC/N6443.Develop solutions for complex business problems                            | 40           | 40              | 0             | 20         | 100         | 20         |
| ASC/N6449.Manage data extraction and collection in automotive manufacturing entities | 40           | 40              | 0             | 20         | 100         | 20         |
| ASC/N6450.Prepare and analyse data by using analytical tools                         | 40           | 40              | 0             | 20         | 100         | 20         |
| ASC/N6451.Analyse assembly line data in automotive manufacturing entity              | 40           | 40              | 0             | 20         | 100         | 15         |
| DGT/VSQ/N0104.Employability Skills (120 Hours)                                       | 20           | 30              | -             | -          | 50          | 10         |
| <b>Total</b>                                                                         | <b>230</b>   | <b>220</b>      | <b>0</b>      | <b>100</b> | <b>550</b>  | <b>100</b> |

## Qualification Pack

### Acronyms

|             |                                                 |
|-------------|-------------------------------------------------|
| <b>NOS</b>  | National Occupational Standard(s)               |
| <b>NSQF</b> | National Skills Qualifications Framework        |
| <b>QP</b>   | Qualifications Pack                             |
| <b>TVET</b> | Technical and Vocational Education and Training |
|             |                                                 |



## Qualification Pack

### Glossary

|                                              |                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sector</b>                                | Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.                                                                      |
| <b>Sub-sector</b>                            | Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.                                                                                                                                                                                     |
| <b>Occupation</b>                            | Occupation is a set of job roles, which perform similar/ related set of functions in an industry.                                                                                                                                                                                                |
| <b>Job role</b>                              | Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.                                                                                                                                                                                |
| <b>Occupational Standards (OS)</b>           | OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts. |
| <b>Performance Criteria (PC)</b>             | Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.                                                                                                                                                                    |
| <b>National Occupational Standards (NOS)</b> | NOS are occupational standards which apply uniquely in the Indian context.                                                                                                                                                                                                                       |
| <b>Qualifications Pack (QP)</b>              | QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.                                                                                                                       |
| <b>Unit Code</b>                             | Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'                                                                                                                                                                                                        |
| <b>Unit Title</b>                            | Unit title gives a clear overall statement about what the incumbent should be able to do.                                                                                                                                                                                                        |
| <b>Description</b>                           | Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.                                                                                                                   |
| <b>Scope</b>                                 | Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.                                                                                            |

## Qualification Pack

|                                         |                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge and Understanding (KU)</b> | Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.                                                                                                                                               |
| <b>Organisational Context</b>           | Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.                                                                                                                                                                       |
| <b>Technical Knowledge</b>              | Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.                                                                                                                                                                                                                                                               |
| <b>Core Skills/ Generic Skills (GS)</b> | Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles. |
| <b>Electives</b>                        | Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.                                                                            |
| <b>Options</b>                          | Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.                                                                                                                                                          |